

Schematic area is on the right hand side; different menu bars and toolbars can be seen on the top of this window.

Since you are opening QUCS for the first time, you can configure the language and appearance settings in the *File*→*Application Settings* menu. It will open the QUCS property editor (Fig. 2) where you can make the desired changes. For the changes to take effect, application must be closed and reopened via *File*→*Exit* menu or *ctrl + Q* shortcut.

Create new schematic

For setting up schematics, let us create a new project. Either press the *New* button above the projects folder or use the menu entry *Project*→*New Project* and enter the new project name. After creating the new project, QUCS shifts to *Content* tab where you can find all the data related to the project, i.e. your schematics, the VHDL files, data display pages, datasheets as well as any other data (such as datasheets). Right hand side is the *untitled* empty schematic (Fig. 3).

To edit the schematic, select the *Components* tab, it shows the lumped components (such as resistors and capacitors), sources (such as DC and AC sources), transmission lines (such as microstrip, coaxial cable and twisted pair), non-linear components (such as ideal op-amp, transistors), digital components (flip-flops), file components (Touchstone files, SPICE files), simulations (AC or DC analysis), diagrams (cartesian or polar

plot) and paintings (texts, arrows and circles) (Fig. 4). Start working on the schematic by placing the components on the schematic area by dragging and

dropping the desired components. You can also click a component once and then move the mouse cursor onto the schematic to click again and put it on that position. Also, you can right click during the mouse movement in order to rotate the components at their final positions.

Design and analysis

For better understanding of the tool, let us take an example of a simple voltage divider circuit and perform a DC steady-state analysis to compute node voltage and branch currents of the circuit.

Fig. 5 shows various components of the circuit placed in the schematic area using the above-mentioned procedure. These components can be connected using a wiring tool, and wiring mode can be enabled by clicking the wiring icon or by *ctrl + E* shortcut. Left click on the component's ports (small red circles) to start a wire and click on the second port to finish the wire. If you want to change the orientation of the wire, right click on it or press *ESC* key to leave the wiring mode.

For any simulation, a reference potential is required (for the nodal analysis). The ground symbol can be found in the *Components* tab in the lumped components category, or you can select

the same by using *ctrl + G* shortcut. Also, you need to place the 'simulation' block in the schematic for each simulation. For the current simulation,

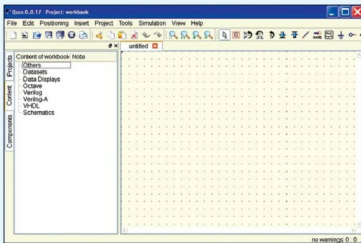


Fig. 3: Untitled empty schematic

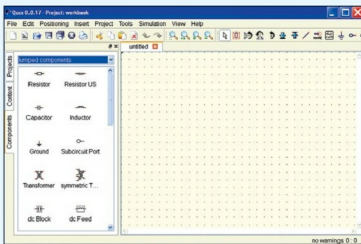


Fig. 4: The components tab window displaying sources, transmission lines, non-linear components, etc

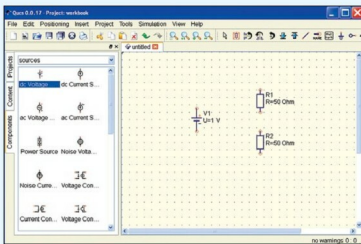


Fig. 5: Various components of the circuit placed in the schematic area